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Prevalence of hepatitis B virus, hepatitis C virus, and human immunodeficiency virus among blood donors of Mekelle blood bank, Northern Ethiopia: A three-year retrospective study

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Blood transfusion services are a vital and integral part of modern healthcare services. However, the risk of transfusion transmittable infections (TTI) has been a major handicap. Therefore, this study was aimed at determining the prevalence of hepatitis B virus (HBV), hepatitis C virus (HCV), and human immunodeficiency virus (HIV) among blood donors. A retrospective study was conducted to collect data about the blood donors who consecutively donated blood from October 2011 to 2014. A three-year retrospective study was conducted in Mekelle Blood Bank. A data abstraction format was used to collect the sociodemographic and clinical data, and the prevalence of HBV, HCV, and HIV was determined. Data were analyzed using STATA version 10 analytical software. *P* value less than 0.05 was considered significant in all the analyses. A total of 10 728 blood donors, median (interquartile range) of age 30 (23-45) years and 3750 (34.9%) males were enrolled in this study. Of the participants 407(3.79%), 143(1.33%), and 111(1.03%) blood donors were positive for HBV, HCV, and HIV, respectively. HBV-HIV coinfections were found 10 (1.93%) blood donors, followed by HBV-HCV and HIV-HCV. A significant association between sex and marital status with HBV and HIV infection was found. However, significant association of HCV was observed among sex ($X^2 = 33.18$, $P < 0.001$) and occupational ($X^2 = 84.33$, $P < 0.001$). A significant percentage of HBV, HCV, and HIV among blood donors was observed. To select a donor and collect safe blood risk factors exposing blood donor should be studied, and community-based prevalence studies on TTI are also required.

KEYWORDS
blood donor, hepatitis B virus (HBV), hepatitis C virus (HCV), human immunodeficiency virus (HIV), prevalence

1 | INTRODUCTION

Blood transfusion is a life-saving intervention, which is an essential therapeutic act that saves millions of lives. It was projected that approximately 90-100 million blood units are collected in the globe

Abbreviations: Ag/Ab, Antigen/antibody; HBC, Hepatitis B virus; HBsAg, Hepatitis B surface antigen; HCV, Hepatitis C virus; HIV, Human immunodeficiency virus; TTI, Transfusion transmittable infections.

each year.[1] In 2005, World Health Organization members of state signed a document that reveal them to the stipulation of safe and adequate blood and blood products to patients.[2]

However, in spite of the benefits of blood transfusion, each transfused patient is at risk of acquiring transfusion transmissible infections, predominantly the human immunodeficiency virus (HIV), hepatitis B virus (HBV), and hepatitis C virus (HCV). World Health Organization recommended blood for transfusion might be screened for HIV, HBV, HCV, and syphilis. In selected countries, depending on epidemiological evidence, additional screening should be also done for the following infections: malaria, Chagas disease, human cytomegalovirus, and human T-cell lymphotropic viruses.[2] HIV, HBV, and HCV are of particular apprehension because of their prolonged infectivity and carrier state, and also cause various devastating disorders, which might ultimately be lethal.[3] Transfusion of infected blood is the cause of 5% to 10% of HIV infections in Sub-Saharan Africa.[4]

The estimate for overall mean risks of becoming infected with HBV, HCV, and HIV from blood transfusion in developed countries is 1/1 700 000 for HBV, 1/6 500 000 for HCV, and 1/2 600 000 for HIV.[5] Whereas in the in Sub-Saharan Africa they were 4.3, 2.5, and 1, respectively, in infections per 1000 units.[6] Therefore, to improve blood safety in African countries, World Health Organization has implemented an approach for Africa, which aims to carry out a national policy for each African nations by 2012, to guarantee a safe and adequate blood supply. This policy includes the conscription of voluntary, regular blood donors, screening for all major transfusion transmissible infections in a quality controlled approach, proper infrastructure, and trained personnel.[7]

In the developed countries, transfusion is most frequently used to support for complex and advanced surgeries such as open-heart surgery, organ transplantation, and advance trauma care although in Sub-Saharan Africa, it is mainly used to treat anemic patients from infections such as malnutrition, malaria, hookworm, and major surgeries apart from acute blood loss from road accidents.[8]

While the developing countries like Africa and Asia, where nationwide blood transfusion services and strategies, trained personnel, adequate infrastructure, and financial resources are inadequate,[9] there is a wide spectrum of bloodborne infections, which can be transmitted through the blood of apparently healthy and asymptomatic blood donors.

However, blood safety remains an issue of major concern in transfusion remedy in Ethiopia where national blood transfusion services and policies, appropriate infrastructure, trained personnel, and financial resources are inadequate. This is aggravated by the lack of comprehensive and systematic screening of donated blood for transfusion transmissible agents. Other main transfusion transmissible infections such as Hepatitis B and C, malaria, and syphilis are not routinely screened. As a result, some of the blood being transfused is likely to contain unscreened pathogens.[10]

Several studies have been carried out to assess the prevalence of HBV, HCV, and HIV, among blood donors in developing and developed countries. This became very crucial because the HIV epidemic appears to be worse in Sub-Saharan Africa.[11] Hence, there is an urgent need for continuous monitoring of the seroprevalence of HBV, HCV, and HIV, which cause life-intimidating infections among blood donors in the study area. This study aimed to determine the Seroprevalence of HBV, HCV, and HIV among blood donors who visit the Mekelle blood bank for three years. This will serve as a baseline data in the monitoring of the prevalence of these transfusion transmissible infections among the blood donor. This will help in formulating policies for improving blood transfusion safety.

2 | METHODS

2.1 | Study setting and study participants

The study was carried out in Mekelle Blood Bank. The blood bank was found in Mekelle city, which is 783 km to the North from Addis Ababa (capital city of Ethiopia). The blood bank participates in activities mainly blood collection from donors and screening of the donated blood for bloodborne pathogens, saving lives of patients who need blood coming from the region and neighboring regions. The blood bank will collect and screen the donated blood by strictly following the national blood screening algorithms. As per the algorithm, healthy donors will be preliminarily selected after asking past medical history and behavior of the donors, and the blood will be collected in the collection bag. The collected blood was screened for TTI.

Laboratory diagnosis of HBV was done by targeting HBsAg using Hepanostika HBsAg enzyme-linked immunosorbent assay (Murex Biotech Ltd, Dartford, UK). In addition to this, serum was also screened for Murex anti-HCV IgG antibodies (DiaSorin, Italy) using enzyme-linked immunosorbent assay. Moreover, diagnosis of donor's serum for HIV was done using Vironostika HIV Uni-Form II Ag/Ab (Bio-Merieux, Boxtel, The Netherlands). All tests were done following the manufacturer's instructions.

2.2 | Study design

A retrospective study was conducted to collect data about the blood donors who consecutively donated blood from October 2011 to 2014. Those blood donors' data with incomplete registered information were excluded from the study. With this, a total of 10 728 blood donor records from the blood bank registration book within the study period were reviewed.

2.3 | Data collection and analysis

A data abstraction format was prepared, and both sociodemographic (like age, sex, marital status, and occupation) and clinical data (like donation type, blood group, HBV status, HCV status, and HIV status) were collected from the blood bank registration book. Data were checked for completeness and accuracy and entered into and analyzed using STATA version 10 analytical software (Statacorp, USA). The presence of association of independent variables with

HBV, HCV, and HIV status was evaluated using chi-square. *P* value less than 0.05 was considered significant in all the analyses.

3 | RESULTS

3.1 | Characteristics of participants

A total of 10 728 blood donors, median (interquartile range) of age 30 (23-45) years, and 3750 (34.9%) males were enrolled in this study. The majority of the blood donors donated blood type O: 3867 (36.1%) followed by B: 3005 (28%), A: 2146 (20%), and AB: 1710 (15.9%). Of the donors, those with Rh-positive blood were only 751 (7%). in total, 4426 (41.3%) blood donors were giving their blood for replacing the blood given for their relatives. Majority of blood donors were students 4292 (40%). From the blood donors who donated their blood during the study period 4655 (43.4%) were single followed by married and divorced (Table 1).

3.2 | Seroprevalence of HBV, HCV, and HIV

All the blood of the study participants were screened for HBV, HCV, and HIV. From the 10 728 study participants, 407(3.79%), 143 (1.33%), and 111(1.03%) blood donors were positive for HBV, HCV, and HIV, respectively. We tried to cross tabulate the coinfections in blood donors. A total of 10 of 518 (1.93%) were HBV-HIV coinfected, followed by HBV-HCV with positivity of 4 of 550 (0.73%) and then by HIV-HCV coinfections of 1 of 254 (0.39%). But, we did not find any blood donor with triple infections (HBV-HBV-HCV).

TABLE 1 Characteristics of the study participants

Variables	Frequency, N (%)
Sex	
Female	6978 (65.1)
Male	3750 (34.9)
Age, y	
18-23	1395 (13)
24-29	2253 (21)
30-35	2145 (20)
36-41	2145 (20)
42-47	1931 (18)
48-53	859 (8)
Marital status	
Married	4563 (42.6)
Single	4655 (43.4)
Divorced	1510 (14)
Occupation	
Student	4292 (40)
Civil servant	3218 (30)
Private workers	2145 (20)
Farmers	1073 (10)
Donor type	
Replacement blood donors	4426 (41.3)
Voluntary blood donors	6302 (58.7)

Of the blood donors who were positive for HBV, 356 (87.47%) were male and 225 (55.28%) were voluntary donors. From the blood donors who were positive for HBV infection, 143(35.14%) were in the age range of 18-23 years, followed by the age range 24-29 years with 117 (28.74%) and 30-35 years with 75(18.43%) (Table 2). Of the 407 positive for HBV infections, 219(53.81%) were single and the majority were students, ie, 165 (40.54%). We found a significant association between sex ($X^2 = 513.12$, $P < 0.001$) and marital status ($X^2 = 20.76$, $P < 0.001$) with HBV infection among blood donors.

Of the143 blood donors who were positive for HCV, 102(71.33%) were male, and 81(56.64%) were voluntary blood donors. From the blood donors who were positive for HCV infection, 60(41.96%) were in the age range of 18-23 years, followed by the age range 30-35 years with 27(18.88%) and 24-29 years with 23(16.09%) (Table 2). Of the positive for HCV infections, 73(51.05%) were single, and the majority were students, ie, 62(43.37%). A significant association between sex ($X^2 = 84.33$, $P < 0.001$) and occupation ($X^2 = 33.18$, $P < 0.001$) with HCV infection among blood donors was found.

The blood donors who were positive for HIV were 111, of which 21 (18.92%) were females and 61 (54.96%) were replacement donors (Table 2). From the blood donors who were positive for HIV infection, 36 (32.43%) were in the age range of 18-23 years, followed by the age range 30-35 years and 24-29 years, with 23 (20.72%) in each. Of the positive for HIV infections, 58(52.25%) were single, and the majority were students, ie, 49(44.14%). A significant association between sex ($X^2 = 104.96$, $P < 0.001$) and marital status ($X^2 = 11.89$, $P < 0.001$) with HIV infection among blood donors was observed.

4 | DISCUSSION

It is known that blood transfusion has been saving millions of lives. The risk of getting infected from microorganisms found in blood and products is one of the most important challenges from blood transfusion. Screening and documentation of such evidence through retrospective studies will have input for indicating the status of volunteer blood donors.

The prevalence of transfusion transmissible infections (TTI) is different in high-income, middle-income, and low -income countries. In this study, we found 3.79%, 1.33%, and 1.03% of the donated blood were positive for HBV, HCV, and HIV respectively. Moreover, we found a significant association of age and marital status with the risk of HBV and HIV infections, and age and occupation with the risk of HCV infection. The finding of this study is in line with the global report about the burden of TTI from low-income countries, but higher than the reports from middle- and high-income countries.[12] This could be explained by the global report on population prevalence of the HBV, HCV, and HIV infections.[13] Compared with Western countries, in a systematic review done by Hahne et al,[14] the prevalence of HBsAg from 24 countries was ranging from 0.0% to

TABLE 2 Association of variables of HBV, HCV, and HIV with risk factors

	HBV		HCV		HIV	
	Positive		Positive		Positive	
Variables	N (%)	X^2 (*P* value)	N (%)	X^2 (*P* value)	N (%)	X^2 (*P* value)
Sex						
Female	51 (12.53)	513.12 ($P < 0.001$)	41 (28.67)	84.33 ($P < 0.001$)	21 (18.92)	104.96 ($P < 0.001$)
Male	356 (87.47)		102 (71.33)		90 (81.08)	
Age, y						
18-23	143 (35.14)	25.23 ($P > 0.25$)	60 (41.96)	9.37 ($P > 0.5$)	36 (32.43)	15.64 ($P > 0.5$)
24-29	117 (28.74)		23 (16.09)		23 (20.72)	
30-35	75 (18.43)		27 (18.88)		23 (20.72)	
36-41	39 (9.58)		21 (14.69)		16 (14.41)	
42-47	28 (6.88)		8 (5.59)		6 (5.41)	
48-60	5 (1.23)		4 (2.79)		7 (6.31)	
Marital status						
Married	132 (32.43)	20.76 ($P < 0.001$)	45 (31.47)	7.33 ($P > 0.025$)	30 (27.03)	11.89 ($P < 0.001$)
Single	219 (53.81)		73 (51.05)		58 (52.25)	
Divorced	56 (13.76)		25 (17.48)		23 (20.72)	
Occupation						
Student	165 (40.54)	5.58 ($P > 0.1$)	62 (43.37)	33.18 ($P < 0.001$)	49 (44.14)	4.32 ($P > 0.1$)
Civil servant	108 (26.54)		37 (25.86)		28 (25.22)	
Private workers	81 (19.90)		12 (8.39)		18 (16.22)	
Farmers	53 (13.02)		32 (22.38)		16 (14.42)	
Type of donation						
Replacement	182 (44.72)	2.09 ($P > 0.1$)	62 (43.36)	0.26 ($P > 0.1$)	61 (54.96)	8.68 ($P > 0.05$)
Voluntary	225 (55.28)		81 (56.64)		50 (45.04)	

HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus.

5.2%, and the prevalence of anti-HCV antibody from 23 countries was from 0.02% to 3.3%.

The findings are relatively higher compared with a 10-year systematic review done in China[15] and a study conducted in Jordan.[16] In the review conducted from China, they found age and marital status to be significantly associated with all the infection, but in the study conducted by Jordan only age was associated. However, similar findings on HBV were found from studies conducted in Saudi Arabia, with age, marital status, educational level, occupation, and family history being significantly associated with HBV infection.[17]

Except few countries, the prevalence of this study is similar to reports from the developing countries. In one study conducted in Brazil, we found relatively lower prevalence of HBV compared with our study. This lower prevalence is associated with the incorporation of the vaccination program for HBV in the general population.[18] However similar findings were reported from studies conducted in India although alcoholism, smoking, tattooing, ear piercing, and visiting barber shop were the risk factors for HBV infection among blood donors.[19]

The prevalence of HBV among blood donors in Africa ranges from 0.007% of research from South Africa[20] and higher prevalence of 18.96% from Niger.[21] Although the prevalence of HCV is lower in most African countries when compared with HBV, higher prevalence of 6.89% was reported in one study from Ivory Coast.[21,22] However the prevalence of HIV among blood donor ranged from 0.00% in Egypt[23] to 4.5% in one study from of Burkina Faso.[24] The difference in the prevalence could be explained by the difference in the population prevalence of the infection in the respective countries.

In Ethiopia, we found six studies conducted among blood donors. Our findings on the prevalence of HBV and HCV is higher than the study conducted by Alemeshet et al,[25] but is lower than studies conducted in Bahdar,[26] Gonder[27] and Mekelle.[28] Similar to our studies, significant associations of HBV and HCV with age and sex,[25] occupation, first-time donors, and age[27] were found.

The finding of our study is in line with the national prevalence of HIV[29] and studies conducted at Jimma Hospital, and unlike our study, the occupation and type of donors were significantly associated with the prevalence among blood donors.[30] However, our findings are lower than reports from Alemeshet et al,[25] Dessie et al,[26] and Tessema et al.[27] Age,[25] sex,[24,27] first-time donors, housewives, merchants, soldiers, drivers, and construction workers[27] were significantly associated with our study.

A higher coinfection of HBV-HIV followed by HBV-HCV and HIV-HCV was found. However, triple infections were not reported. Our findings are in agreement with studies conducted in Cameron.[31] However, the findings of HBV-HIV confections are relatively lower than studies conducted by Kossovo[32] and Ghana.[33]

5 | CONCLUSION

We found a significant percentage of the blood donors harboring transfusion transmissible infections, with higher prevalence of HBV followed by HCV and HIV. Besides, coinfections of HBV-HIV were higher compared with HCV-HIV and HBV-HCV. We found a significant association between sex and marital status with HBV and HIV infections. However, in addition to age, occupation was also significantly associated with HCV infections. To select donor and collect safe blood, the risk factors exposing blood donor for infections should be studied and community-based prevalence studies on the TTI are also required.

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CONFLICTS OF INTEREST

The authors have no conflict of interests to declare.

AUTHOR CONTRIBUTIONS

NS and MG conceived and designed the study. NS, GB, GZ, GT, carried out coordination of the study. GA, HH, TE and AM participated during data collection and data entry. NS and MG carried out statistical analysis and developed manuscript. All authors read and approved the final manuscript.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

This study was evaluated and approved by the Research Ethics Review Committee (RERC) of College of Health Sciences, Mekelle University. Moreover, letter of cooperation and permission to access Mekelle blood bank registration book was obtained from the blood bank management bodies. The consent involves permission to disseminate the findings of the study through the scientific workshop and publish in reputable journals

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REFERENCES

1. WHO. Global strategic plan, 2008-2015. http://www.who.int/bloodsafety/StrategicPlan2008-2015AccessSafeBloodTransfusion.pdf. Accessed November 10, 2015.
2. WHO. Screening donated blood for transfusion transmissible infections: recommendations 2010. http://www.who.int/bloodsafety/ScreeningDonatedBloodforTransfusion.pdf. Accessed November 10, 2015.
3. Wallace RB. *Maxey-Rosenau-Last Public Health and Preventive Medicine: Fifteenth Edition*. New York, NY: McGraw-Hill; 2007:191-221.
4. Lackritz EM. Prevention of HIV transmission by blood transfusion in the developing world: achievements and continuing challenges. *AIDS*. 1998;12(Suppl A):S81-S86.
5. Traineau R, Elghouzzi MH, Bierling P. Update on infectious risks associated with blood products. *Rev Prat*. 2009;59:86-89.
6. Jayaraman S, Chalabi Z, Perel P, Guerriero C, Roberts I. The risk of transfusion-transmitted infections in Sub-Saharan Africa. *Transfusion*. 2010;50:433-442.
7. WHO IRIS Blood safety: a strategy for the African region. 2001. http://apps.who.int/iris/handle/10665/95734. Accessed November 10, 2015.
8. Tapko JB, Mainuka P, Diarra-Nama AJ Status of blood safety in the WHO African region: report of the 2006 survey. 2006. http://www.afro.who.int/index.php?option=com_docman&task=doc_download&gid=3835. Accessed November 10, 2015.
9. Nagalo MB, Sanou M, Bisseye C, et al. Seroprevalence of human immunodeficiency virus, hepatitis B and C viruses and syphilis among blood donors in Koudougou (Burkina Faso) in 2009. *Blood Transfus*. 2011;9:419-424.
10. Allain JP, Stramer SL, Carneiro-Proietti ABF, et al. Transfusion-transmitted infectious diseases. *Biologicals*. 2009;37:71-77.
11. UNAIDS. Global report: UNAIDS report on the global AIDS epidemic. 2013. http://www.unaids.org/sites/default/files/media_asset/UNAIDS_Global_Report_2013_en_1.pdf. Accessed November 10, 2015.
12. WHO. Blood safety and availability. 2015. file:///C:/Users/Vostro%202520/Desktop/HBV%20maniscript_discussion/WHO%20fact%20sheet%20globe.htm. Accessed October 24, 2015.
13. WHO. HIV and hepatitis co-infections. 2015. http://www.who.int/hiv/topics/hepatitis/en/. Accessed October 24, 2015.
14. Hahné SJ, Veldhuijzen IK, Wiessing L, et al. Infection with hepatitis B and C virus in Europe:a systematic review of prevalence and cost-effectiveness of screening. *BMC Infect Dis*. 2013;13:181.
15. Li C, Xiao X, Yin H, et al. Prevalence and prevalence trends of transfusion transmissible infections among blood donors at four Chinese regional blood centers between 2000 and 2010. *J Transl Med*. 2012;10:176.
16. Al-Gani Fathi Abed. Prevalence of HBV, HCV and HIV-1, 2 infections among blood donors in Prince Rashed Ben Al-Hassan Hospital in North Region of Jordan. *Int J Biol Med Res*. 2011;2(4):912-916.
17. El Beltagy KE, Al Balawi IA, Almuneef M, Memish ZA. Prevalence of hepatitis B virus markers among blood donors in a tertiary hospital in Tabuk, Northwestern Saudi Arabia. *Int J Infect Dis*. 2008;12:495-499.
18. Lindenberg ASC, Motta-Castro ARC, Puga MA, et al. Decrease in hepatitis B prevalence among blood donors in Central-West Brazil. *J Venom Anim Toxins Incl Trop Dis*. 2013;19:7.
19. Lavanya V, Viswanathan T, Arul Sheeba Malar S, Malarvizhi A, Moorthy K. Prevalence of hepatitis B virus infection among blood donors with antibodies to hepatitis B core antigen. *Int. J. Med. Med. Sci*. 2012;4(6):128-137.
20. Vermeulen M, Lelie N, Sykes W, et al. Impact of individual-donation nucleic acid testing on risk of human immunodeficiency virus, hepatitis B virus, and hepatitis C virus transmission by blood transfusion in South Africa. *Transfusion*. 2009;49:1115-1125.
21. Tagny CT, Diarra A, Yahaya R, et al. Characteristics of blood donors and donated blood in Sub-Saharan Francophone Africa. *Transfusion*. 2009;49:1592-1599.
22. Bloch EM, Vermeulen M, Murphy E. Blood Transfusion Safety in Africa: A Literature Review of Infectious Disease and Organizational Challenges. *Transfus Med Rev*. 2012;26(2):164-180.
23. El-Gilany AH, El-Fedawy S. Bloodborne infections among student voluntary blood donors in Mansoura University, Egypt. *East Mediterr Health J*. 2006;12:742-748.
24. Collenberg E, Ouedraogo T, Ganamé J, et al. Seroprevalence of six different viruses among pregnant women and blood donors in rural and urban Burkina Faso: A comparative analysis. *J Med Virol*. 2006;78:683-692.

25. Yami A, Alemseged F, Hassen A. Hepatitis B and C viruses infections and their association with human immunodeficiency virus: a cross-sectional study among blood donors in Ethiopia. *Ethiop J Health Sci*. 2011;21(1):67-75.
26. Dessie A, Abera B, Wale F. Seroprevalence of major blood-borne infections among blood donors at Felege Hiwot referral hospital, northwest Ethiopia. *Ethiop J Health Dev*. 2007;21(1):68-69.
27. Tessema B, Yismaw G, Kassu A, et al. Seroprevalence of HIV, HBV, HCV and syphilis infections among blood donors at Gondar University Teaching Hospital, Northwest Ethiopia: declining trends over a period of five years. *BMC Infet Dis.*. 2010;10:111.
28. Gelaw B, Mengistu Y. The prevalence of HBV, HCV and malaria parasites among blood donors in Amhara and Tigray regional states. *Ethiop.J.Health Dev*. 2007;22(1):3-7.
29. Federal HIV/AIDS prevention and control office report 2015. HIV Prevalence level shows a drastic Fall in Ethiopia. http://www.hapco.gov.et/index.php/media-room/news/item/12-hiv-prevalence-level-shows-a-drastic-fall-in-ethiopia. Accessed on 10 Nov 2015.
30. Gezahegn M, Woldemichael K, Godesso A. HIV sero-prevalence trend among blood donors in Jimma University Specialized Hospital, Southwest Ethiopia. *Ethiop J Health Sci*. 2012;22(1):37-43.
31. Fouelifack Ymele F, Keugoung B, Fouedjio JH, Kouam N, Mendibi S, Dongtsa Mabou J. High rates of Hepatitis B and C and HIV Infections among blood donors in Cameroon: A proposed blood screening algorithm for blood donors in resource-limited settings. *J Blood Transfus*. 2012;2012:1-7. https://doi.org/10.1155/2012/458372.
32. Fejza H, Telaku S. Prevalence of HBV and HCV among blood donors in Kosovo. *Virol J*. 2009;6(21):1-5.
33. Nkrumah B, Owusu M, Frempong HO, Averu P. Infections among blood donors from rural Ghana. *Ghana Med J*. 2011;45(3):97-100.

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